B.M.S COLLEGE OF ENGINEERING BENGALURU
Autonomous Institute, Affiliated to VTU

SPC AAT Report on

TITLE

Bill calculator

Bachelor of Engineering
in
Artificial Intelligence and Machine Learning

Submitted by:

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2. Bhagya

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2025-2026

B.M.S COLLEGE OF ENGINEERING

DEPARTMENT OF AIRTIFIACAL

DECLARATION

We, Aditi R Adrushappanavar and Bhagya, students of first Semester, B.E, Department of AIML, BMS College of Engineering, Bangalore, hereby declare that, this AAT Project entitled "Bill Calculator" has been carried out in Department of CSE, BMS College of Engineering, Bangalore during the academic semester September 2025 to January 2026. We also declare that to the best of our knowledge and belief, the AAT Project report is not part of any other report by any other students.

Student Name

1.Aditi R Adrushappanavar

2. Bhagya

BMS COLLEGE OF ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the AAT Project titled "**Bill Calculator**" has been carried Aditi.R.Adrushappanavar (1BM25AI355-T) and Bhagya (1BM25AI240-T) out during the academic year 2025-2026.

Signature of the Faculty in Charge

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INTRODUCTION

This C program is designed to calculate the final bill amount for a simple billing system by considering quantity, price, discount, and tax. It demonstrates the practical use of basic C programming concepts such as variables, data types, input/output functions, arithmetic operations, and formatted output. The program begins by including the standard input-output header file <stdio.h>, which allows the use of functions like printf() and scanf() for interaction with the user.

The program takes essential inputs from the user, including the quantity of items sold, the price per item, the discount percentage, and the tax percentage. Using these inputs, it first calculates the total cost before applying any discount or tax. Then, it computes the discount amount based on the given discount percentage and applies tax on the discounted total. Finally, the program calculates the final bill amount by subtracting the discount and adding the tax.

This code helps students understand real-life applications of C programming in areas such as shopping systems and billing software. It also teaches how to perform step-by-step calculations and display results in a clear and organized manner.

ALGORITHM

1. Start
2. Declare integer variable quantity.
3. Declare float variables price, discountPercent, taxPercent.
4. Declare float variables total, discountAmount, taxAmount, finalAmount.
5. Read the value of quantity.
6. Read the value of the price.
7. Read the value of discountPercent.
8. Read the value of taxPercent.

total = quantity × price

10. Compute discount amount:

    discountAmount = (discountPercent / 100) × total

11. Compute tax amount:

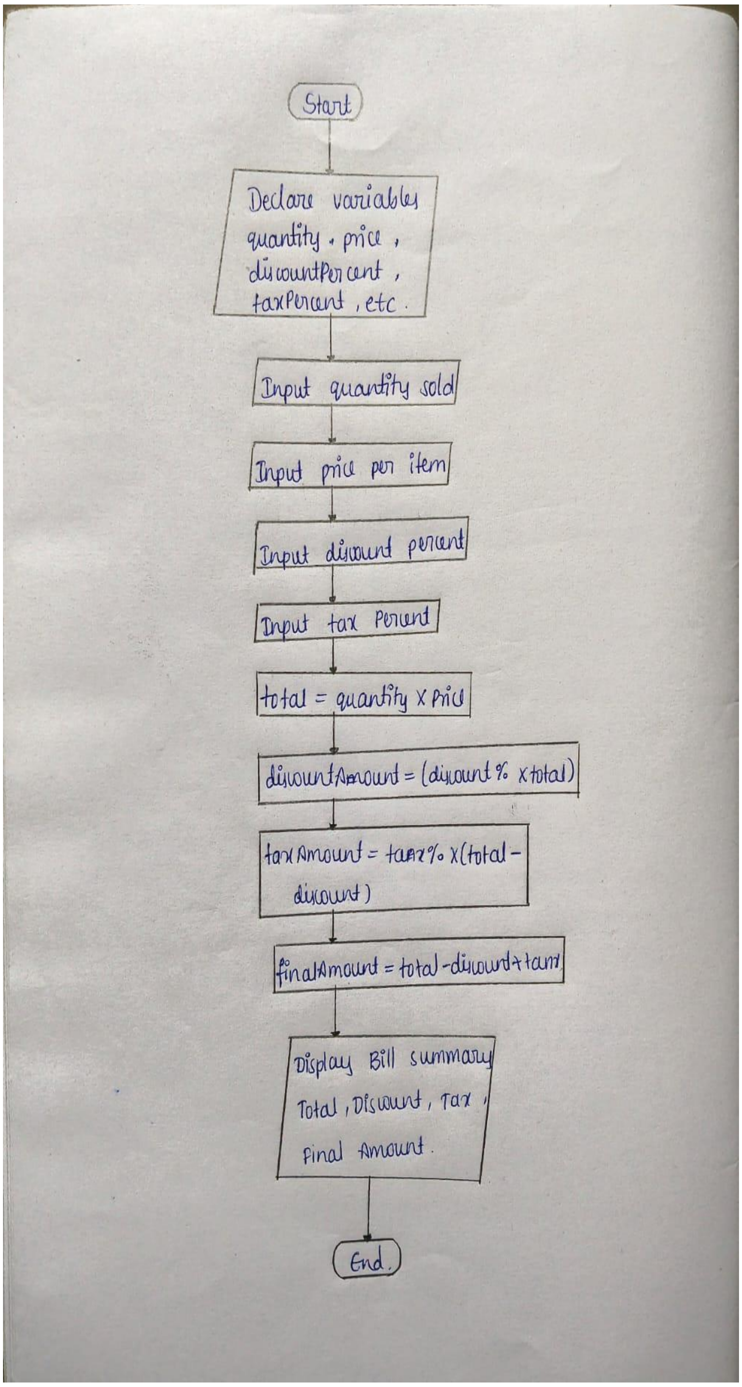
    taxAmount = (taxPercent / 100) × (total − discountAmount)

12. Compute final bill amount:

    finalAmount = total − discountAmount + taxAmount

13. Display total amount, discount amount, tax amount, and final bill amount.
14. Stop

FLOWCHART



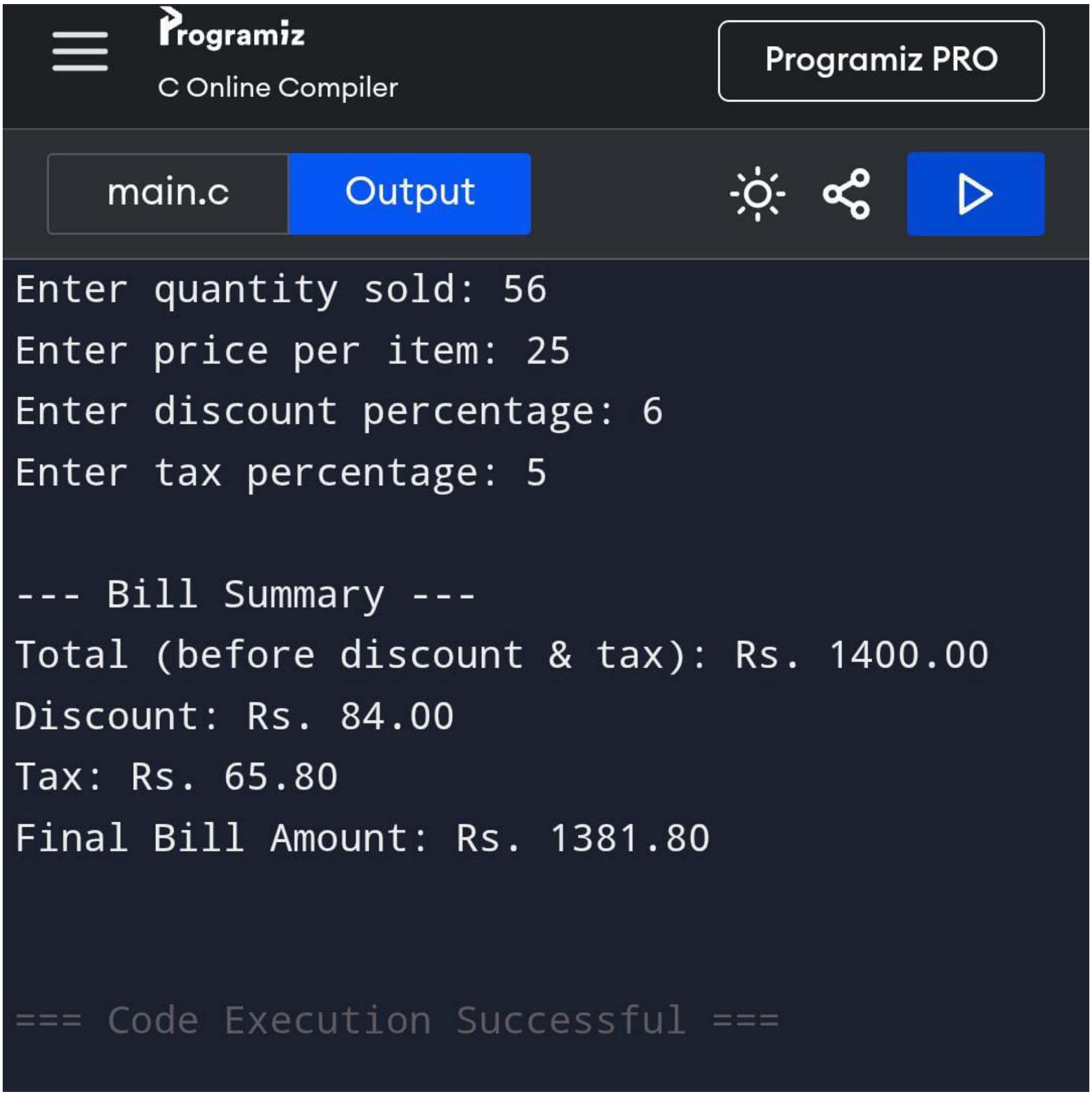
Start
Declare variables quantity, price, discountPercent, taxPercent, etc.
Input quantity sold
Input price per item
Input discount percent
Input tax Percent
total = quantity X Price
discountAmount = (discount % X total)
taxAmount = tax% X (total - discount)
finalAmount = total - discount + tax
Display Bill summary Total, Discount, Tax, Final Amount.
End.

SOURCE CODE

```
#include<stdio.h>
int main ()
{
int quantity;
float price, discountPercent, taxPercent;
float total, discountAmount, taxAmount, finalAmount;
//Input values
Printf("Enter quantity sold:");
Scanf("%d",&quantity);
Printf("Enter price per item:");
Scanf("%f",&price)
Printf("Enter discount percentage:");
Scanf("%f", &discountPercent);
Printf("Enter tax percentage:");
Scanf("%f", &taxPercent);
//Calculate total before discount and tax
total=quantity*price;
//Calculate discount and tax
discountAmount=(discountPercent/100) *total;
taxAmount=(taxPercent/100) *(total-discountAmount);
//Final amount
finalAmount=total-discountAmount+taxAmount;
//Display breakdown and final amount
printf("\n---Bill Summary---\n");
printf("Total (before discount &tax): Rs.%.2f\n",total);
printf("Discount: Rs.%.2f\n",discountAmount);
```

```
printf(“Tax: Rs.%.2f\n”,taxAmount);
printf(“Final Bill Amount: Rs.%.2f\n”,finalAmount);
Return 0;
}
```

RESULTS



```
Enter quantity sold: 25
Enter price per item: 13
Enter discount percentage: 4
Enter tax percentage: 3

--- Bill Summary ---
Total (before discount & tax): Rs. 325.00
Discount: Rs. 13.00
Tax: Rs. 9.36
Final Bill Amount: Rs. 321.36


=== Code Execution Successful ===
```

REFERENCES

1. The Complete Reference for C
2. Computer Fundamentals and Programming in C
3. C Programming : A Modern Approach
4. https://youtu.be/5QLxkgnl1Nc?si=0UR6PLhYSBun2tOX
5. https://youtu.be/kCDZArDdPRo?si=ED6h1R7mwHm32fc9